PGR 111, tasks for week 04

Task 1 - Concepts & Terminology

Explain the following terminology and database concepts:

- Cartesian product
- (INNER) JOIN
- LEFT (OUTER) JOIN
- RIGHT (OUTER) JOIN

NB: Keep your answers. They might become helpful later in the semester. :-)

Task 2 - SELECT queries

Use the MySQL sample database ("schema") called **world**. (Same as we have been working with earlier.)

Write SQL queries that perform the following:

1. Extract all cities *beginning **and** ending* with 'o' (uppercase first, small at the back) and their population figures. Sort them alphabetically. (Tip: Wildcard. And *no* join in this one.)

2. Task 1 gave many city names, but where are these cities located? We want to include the name of the country in which the cities are located. Thus, the columns will be country names, city names and city populations. We want the answer sorted alphabetically by country, then by city.
   Problem: We must query two tables at once! Rewrite the query from task 1 to a JOIN query, where we include the country table, and extract the names of the countries from it.
   Tip: You need to find the common column (foreign key to primary key connection) for these two tables.

3. Extract all countries (their names). Also have a column showing the names of their capitals (AS Capital), if any. Sort the result by country name. Hint: My answer returns 239 rows - is that the same number of rows that you get? *(Note: The Capital column in country matches the ID column in city.)*

4. Retrieve an overview of all countries (Names) that have at least one city, how many cities they have (AS NumberOfCities) and the average population of these cities (AS AverageCityPopulation).

5. Retrieve all the countries in the world (names, as CountryName) where the population of the country is less than 1000 people. We also want to retrieve the population of the country (AS CountryPop) and which continent the country belongs to. In addition, show names of cities (as CityName) in each country if any, and the population of these cities (as CityPop). We want the result sorted alphabetically by CountryName. (Hint: My answer returns 10 rows - is that the same number of rows that you get?)

6. List the cities of the world located in a country with some form of monarchical system of government. Show the city's name, the city's population, and the country's government form. Sort by the city with the highest population first.

7. Retrieve a list of cities (their names and population) in the world that have at least 8,000,000 inhabitants, and which country and continent they belong to. Sort by the cities' population, largest first. Use aliases (AS) if there are ambiguous column names in the result.

8. Create a query that extracts an overview of the countries in Asia where at least 10 different languages are spoken. The query should contain: Name of country and number of languages. It should be sorted descending by number of languages.

Task 3 – Create your own SELECTs

Regarding the world database, come up with some interesting queries you want to execute. Feel free to use different JOINs, GROUP BY, composite WHERE-clauses (AND/OR), HAVING and using wildcards! :-D

Once you have created some tasks and posted them on the Canvas forum, solve tasks from others with a level of difficulty that suits you. :-)